Yubeen Jeong

Earth, Marine, and Environmental Sciences
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Education

08.2022 – 05.2027 (expected) **University of North Carolina at Chapel Hill**
Ph.D. Marine Sciences (Advisor: Dr. Harvey Seim)

03.2019 – 02.2021 **Pusan National University**
M.S. Oceanography (Advisor: Dr. Young-Heon Jo)

03.2015 – 02.2019 **Pusan National University**
B.S. Oceanography

Professional Experience

08.2022 – present **Research/Teaching Assistant**
University of North Carolina at Chapel Hill

03.2021 – 07.2022 **Associate Researcher**
Pusan National University

Research Interests

- Western boundary current dynamics and long-term variability
- Current-topography interactions and boundary current separation
- Mesoscale processes and air-sea interactions
- Observational oceanography integrating in situ measurements and satellite remote sensing

Manuscripts in Preparation

[1] **Jeong, Y.**, Seim, H., Muglia, M., Taylor, T. (in prep). Extraction and Evaluation of Gulf Stream Parameters from Long-Term High-Frequency Radar Observations

[2] **Jeong, Y.**, et al. (in prep). Observed Equatorward Shift Events of the Gulf Stream near Cape Hatteras and Their Atmospheric Forcing

[3] **Jeong, Y.**, et al. (in prep). Repeated Recovery from Topographic Perturbations Preserves Gulf Stream Path Coherence before Separation

Publications (peer reviewed)

[1] Speciale, E. V., Lim, P., **Jeong, Y.**, Cook, C. C., Cohen, N. R., McClure, W., Schnetzer, A., Till, C. P., Till, R., Marchetti, A. (2026). Mixotrophy as an alternative strategy under iron-limited upwelling conditions, ISME Communications. https://doi.org/10.1093/ismeco/ycag290

[2] Kim, D. Y., Kim, D., **Jeong, Y.**, Jo, Y.-H. (2024). Validation of Spatial Boundary of the Ulleung Warm Eddy Using Altimetry. Korean Journal of Remote Sensing, 40(6), 1019-1026. https://doi.org/10.7780/kjrs.2024.40.6.1.12

[3] Yun, M. S., Kang, J. J., **Jeong, Y.**, Jo, Y.-H., Sun, J., Lee, S. H. (2024). Experimental Assessment of Ultraviolet Radiation Impact on the Primary Production of Phytoplankton in the East/Japan Sea. Journal of Marine Science and Engineering,12(8), 1258. doi:https://doi.org/10.3390/jmse12081258

[4] **Jeong, Y.**, Nam, S. H., Kwon, J.-I., Jo, Y.-H., Uppara, U. (2022) Surface Warming Slowdown with Continued Subsurface Warming in the East/Japan Sea Over Recent Decades (2000-2014). Frontiers in Marine Science.9:825368. https://doi.org/10.3389/fmars.2022.825368

[5] Yun, M. S., Kim, Y., **Jeong, Y.**, et al. (2020). Weak response of biological productivity and community structure of phytoplankton to mesoscale eddies in the oligotrophic Philippine Sea. Journal of Geophysical Research: Oceans, 125, e2020JC016436. https://doi.org/10.1029/2020JC016436

[6] **Jeong, Y.**, Hwang, J., Park, J., Jang, C.J., Jo, Y.-H. (2019). Reconstructed 3-D Ocean Temperature Derived from Remotely Sensed Sea Surface Measurements for Mixed Layer Depth Analysis. Remote Sensing, 11(24), 3018; https://doi.org/10.3390/rs11243018

[7] **Jeong, Y.**, Kim, D., Jo, Y.-H., Kim, D.-W., (2019). Interactions of eddies with the Kuroshio Current based on satellite altimeter measurements. Journal of Coastal Research, Special Issue No. 90, pp. 289-293. https://doi.org/10.2112/SI90-036.1

Published Datasets

[1] Speciale, E., Lim, P., **Jeong, Y.**, Cook, C., Cohen, N., McClure, W., Schnetzer, A., Marchetti, A. (2026) Physiological measurements from an iron incubation experiment using upwelled waters in the California Current System during the PUPCYCLE II cruise in May and June 2023. Biological and Chemical Oceanography Data Management Office (BCO-DMO). (Version 1) Version Date 2026-08-12. https://doi.org/10.26008/1912/bco-dmo.994607.1

[2] Speciale, E., Lim, P., **Jeong, Y.**, Cook, C., Cohen, N., McClure, W., Schnetzer, A., Marchetti, A. (2026) Pfam differential expression data for diatoms from an iron incubation experiment aboard the R/V Sally Ride during the PUPCYCLE II cruise within the California Current System in May-June 2023. Biological and Chemical Oceanography Data Management Office (BCO-DMO). (Version 1) Version Date 2026-09-13. https://doi.org/10.26008/1912/bco-dmo.1004674.1

[3] Speciale, E., Lim, P., **Jeong, Y.**, Cook, C., Cohen, N., McClure, W., Schnetzer, A., Marchetti, A. (2026) Metatranscriptomic (RNA-seq) taxonomy data from an iron incubation experiment during the PUPCYCLE II cruise (R/V Sally Ride) within the California Current System in May and June of 2023. Biological and Chemical Oceanography Data Management Office (BCO-DMO). (Version 1) Version Date 2026-08-12. https://doi.org/10.26008/1912/bco-dmo.1003525.1

Presentations

[1] **(Oral) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M., Taylor, T. (2026). Observed Equatorward Shift Events of the Gulf Stream near Cape Hatteras and Their Atmospheric Forcing. 2026 American Geophysical Union (upcoming)

[2] **(Oral) Jeong, Y.** (2026). Multi-Scale Variability and Dynamical Controls of the Gulf Stream from Long-Term High-Frequency Radar Observations. 2026 Physical Oceanography Dissertation Symposium XIV (upcoming)

[3] **(Poster) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M., Taylor, T. (2026). Localized southward shift of the Gulf Stream and their atmospheric drivers observed by long-term high-frequency radar near Cape Hatteras. 2026 Ocean Sciences Meeting

[4] **(Poster)** Muglia, M., Taylor, T., Seim, H., Bane, J., Lowcher, C., **Jeong, Y.** (2026). A twenty-year Gulf Stream meander census off Cape Hatteras, North Carolina, using surface currents from monostatic high-frequency radars. 2026 Ocean Sciences Meeting

[5] **(Poster) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M. (2025). Long-term analysis of Gulf Stream variability at Cape Hatteras using High-Frequency radar surface current observations. 2025 EMES Student Research Symposium

[6] **(Oral) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M., Lowcher, C., Taylor, T. (2025). Long-term analysis of Gulf Stream variability at Cape Hatteras using High-Frequency radar surface current observations. 2025 ASLO Aquatic Sciences Meeting

[7] **(Oral)** Speciale, E., Lim, P., **Jeong, Y.**, Marchetti, A. (2025). Battling Trophic Modes: How Do Mixotrophs Respond to Iron Limitation in Upwelling Zones? 2025 ASLO Aquatic Sciences Meeting

[8] **(Poster) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M. (2024). Long-term analysis of Gulf Stream variability at Cape Hatteras using High-Frequency radar surface current observations. 2024 EMES Student Research Symposium

[9] **(Poster) Jeong, Y.**, Seim, H., Bane, J., Andres, M., Muglia, M. (2024). Long-term analysis of Gulf Stream variability at Cape Hatteras using High-Frequency radar surface current observations. 2024 Ocean Sciences Meeting (OSM)

[10] **(Poster)** Lowcher, C., Muglia, M., Seim, H., Al-Attabi, Z., Taylor, T., Bane, J., **Jeong, Y.** (2024). HF Radar Observations of Gulf Stream Characteristics off of Cape Hatteras, North Carolina. 2024 Ocean Sciences Meeting (OSM)

[11] **(Poster)** Choi, J. G., Lippmann, T. C., Pringle, J. M., Kim, D., **Jeong, Y.**, Jo, Y.-H. (2024). Validity and limitation of wind impulse as index of upwelling. 2024 Ocean Sciences Meeting (OSM)

[12] **(Poster)** Seim, H., Edwards, C., **Jeong, Y.** (2023). Persistent wintertime cascading at the shelfbreak adjacent to a subtropical western boundary current in the NW Atlantic Ocean. International Workshop on Western Boundary Current

[13] **(Oral) Jeong, Y.**, Seim, H. (2023). G1glider's non-pumped CTD Salinity Correction. SouthEast Coastal Ocean Observing Regional Association (SECOORA) hurricane glider tech talk

[14] **(Oral) Jeong, Y.**, Seim, H., Bane, J. (2023). Long-term observation of the Gulf Stream off North Carolina using HF Radar. 2023 EMES PerkinElmer Student Research Symposium

[15] **(Oral) Jeong, Y.**, Nam, S. H., Kwon, J.-I., Jo, Y.-H. (2022). Surface warming slowdown with continued subsurface warming in the East/Japan Sea over the recent decades (2000-2014). 2022 Ocean Sciences Meeting (OSM)

[16] **(Oral) Jeong, Y.**, Nam, S. H., Kwon, J.-I., Jo, Y.-H. (2021). Causes of the Surface Warming Slowdown in the East/Japan Sea. 2021 Asia Oceania Geosciences Society (AOGS)

[17] **(Poster) Jeong, Y.**, Nam, S. H., Kwon, J.-I., Jo, Y.-H. (2021). Surface warming slowdown with continued subsurface warming in the East/Japan Sea over the recent decades (2000-2014). Korean Society of Oceanography (KSO), Spring meeting
(This poster won the best student poster award in the division of physical oceanography at KSO)*

[18] **(Poster) Jeong, Y.**, Jo, Y.-H. (2020). Causes of the Surface Warming Slowdown in the East/Japan Sea. Korean Society of Oceanography (KSO), fall meeting

[19] **(Oral) Jeong, Y.**, Jo, Y.-H. (2020). Causes of the Surface Warming Slowdown in the East/Japan Sea. 2020 Ocean & Atmosphere Science Interdisciplinary Studies (OASIS), oral presentation
(This presentation won the best oral presentation award)*

[20] **(Poster) Jeong, Y.**, Jo, Y.-H. (2020). The processes of surface warming hiatus in the East/Japan Sea. Korean Society of Oceanography (KSO), spring meeting

[21] **(Poster) Jeong, Y.**, Jo, Y.-H. (2019). The processes of surface warming hiatus in the East/Japan Sea. Korean Society of Oceanography (KSO), fall meeting

Teaching Experiences

(Spring 2023) | Graduate Teaching Assistant, MASC 101L, The Marine Environment Laboratory
Led laboratory sessions and guided students through experimental procedures.

(Spring 2023) | Graduate Teaching Assistant, MASC 101, The Marine Environment
Graded assignments and held office hours.

(Spring 2024 - Spring 2025) | Undergraduate mentoring
Mentored undergraduate students in processing Gulf Stream time-series data from the Cape Hatteras region and the Florida Straits.

Invited Talks

2025	Long-term observation of the Gulf Stream off North Carolina using HF Radar. Pusan National University
2024	Long-term observation of the Gulf Stream off North Carolina using HF Radar. Seoul National University

Professional Service

02.2026	**Student Session Moderator, Ocean Sciences Meeting, Glasgow, UK** Session: PS003 Dynamics of the ocean's overturning: past, present, and future

05.2023-present	**Glider Piloting (SECOORA project)** Optimizing waypoint navigation for underwater gliders in dynamic ocean conditions and monitoring glider behavior Providing shore-based support for glider deployment and recovery operations Correcting salinity data from the G1 Glider's non-pumped CTD sensor

Awards, Grants, and Fellowships

2027 (selected)	**Department dissertation completion fellowship** Earth, Marine, and Environmental Sciences Department, UNC Chapel Hill
2026 (selected)	**Selected Participant** Physical Oceanography Dissertation Symposium (PODS XIV)
2022	**Gussenhoven Graduate Student Fellowship ($4000)** Earth, Marine, and Environmental Sciences Department, UNC Chapel Hill
2021	**Korean Society of Oceanography, Spring meeting, Best student poster award** Korean Society of Oceanography (KSO)
2020	**Ocean & Atmosphere Science Interdisciplinary Studies (OASIS) Oral Presentation Grand Prize** Korea Hydrographic and Oceanographic Agency (KHOA)
2020	**PNU Graduate School Academic Award (Excellent Paper Award)** Pusan National University (PNU)
2018	**Scholarship for Academic Excellence** Pusan National University (PNU)
2017	**Scholarship for talented students from leading university promotion project of Busan** National Research Foundation of Korea (NRF)

Research Cruises

05.27.-06.12. 2023	**R/V Sally Ride** CTD operations and drifter deployment (Upwelling Dynamics) California Coast
01.02.-01.12. 2023	**R/V Neil Armstrong** CPIES deployment (NASA SWOT cal/val project) Cape Hatteras (U.S. East Coast)

Field Experiences

03.12.-03.17. 2023 **High-Frequency Radar Repair and Reinstallation**
Outer Banks, NC – HATY, DUCK, JENN sites

03. 2019 **R/V NARA**
Operated CTD instruments and trained undergraduate students in their use.

Professional Training

(08. 2022) North Pacific Marine Science Organization PICES Summer School [Ocean Big Data]

Analysis of observed hourly submesoscale surface current maps

Technical Skills

Programming: MATLAB, Python

Data Analysis: Oceanographic data processing; time-series analysis; scientific visualization

Memberships

American Geophysical Union (AGU)

Asia Oceania Geosciences Society (AOGS)

The Korean Society of Oceanography (KSO)

References

Dr. Harvey Seim

Professor

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University of North Carolina at Chapel Hill

Email: hseim@email.unc.edu

Relationship: Ph.D. Advisor

Dr. John Bane

Research Professor

Department of Earth, Marine, and Environmental Sciences

University of North Carolina at Chapel Hill

Email: jbanejr@gmail.com

Relationship: Dissertation Committee Member

Dr. Magdalena Andres

Senior Scientist

Department of Physical Oceanography

Woods Hole Oceanographic Institution

Email: mandres@whoi.edu

Relationship: Dissertation Committee Member

Dr. Catherine Edwards

Associate Professor

Department of Marine Sciences

Skidaway Institute of Oceanography, University of Georgia

Email: catherine.edwards@skio.uga.edu

Relationship: Dissertation Committee Member

Dr. Mike Muglia

Interim Executive Director, Associate Scientist, Coastal Studies Institute

Research Associate Professor, East Carolina University

Assistant Director for Science & Research, North Carolina Renewable Ocean Energy Program

Email: Mugliam@ecu.edu

Relationship: Research/Project Collaborator